

Data Analytics – Stage Two (2B)

Author: Oluwadoromi Emmanuel

Title: Outlier Detection and Geospatial Anomaly Report for Lagos State 2023 Presidential Election

Introduction

This report presents a two-layered anomaly detection approach applied for Lagos State's 2023 presidential election data. Statistical and geospatial models were used, the analysis reveal polling units (PUs) whose results deviate sharply from both statewide voting trends and local neighborhood patterns. These findings helped reveal several extreme anomalies across diverse LGAs, highlighting potential data inconsistencies or irregular vote patterns warranting further verification.

Data Preparation

- **Data Source:** Lagos State election data with 11,911 polling units across 19 columns
- **Geocoding:** Used Nominatim geocoder to obtain latitude and longitude coordinates for each LGA
- **Key Features Created:**
 - **Voter Turnout:** $(\text{Accredited_Voters} / \text{Registered_Voters}) \times 100$
 - **Party Performance Index (PPI):** Measures how each polling unit's party performance compares to state average
 - **Total Votes per Polling Unit:** Sum of votes for all four major parties (APC, LP, PDP, NNPP)

Outlier Detection Techniques

1. Isolation Forest Algorithm

- Applied on standardized features: Latitude, Longitude, Voter Turnout, and PPI_LP
- Contamination parameter set to 5% (0.05)
- Generates anomaly scores indicating how "isolated" each polling unit is from others

2. **Geospatial Deviation Analysis using Ball Tree**

- Used Haversine distance metric for geographical calculations
- Radius of 1km (1/6371 radians) to define neighbourhood
- For each polling unit, compared its party votes with neighbouring units
- Calculated standardized deviation scores for each party:

$$\text{Deviation} = \frac{\text{Current_PU_Votes} - \text{Neighborhood_Mean}}{\text{Neighborhood_STD}}$$

3. **Final Outlier Scoring**

- Calculated combine score as:

$$\text{Final_Outlier_Score} = \text{Max_Geo_Deviation} - \text{Anomaly_Score}$$

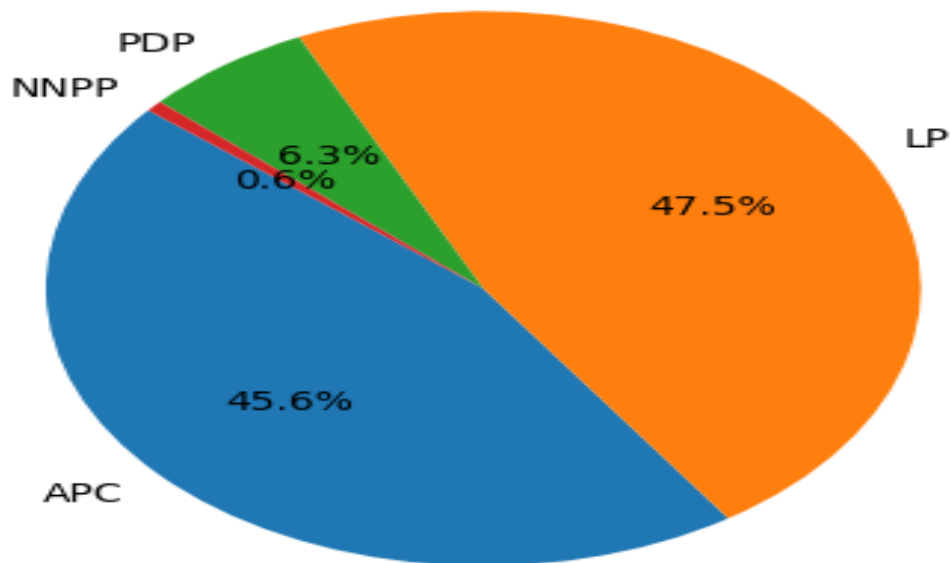
- Higher scores indicate stronger outliers

Summary of Findings

Overall Party Performance in Lagos State

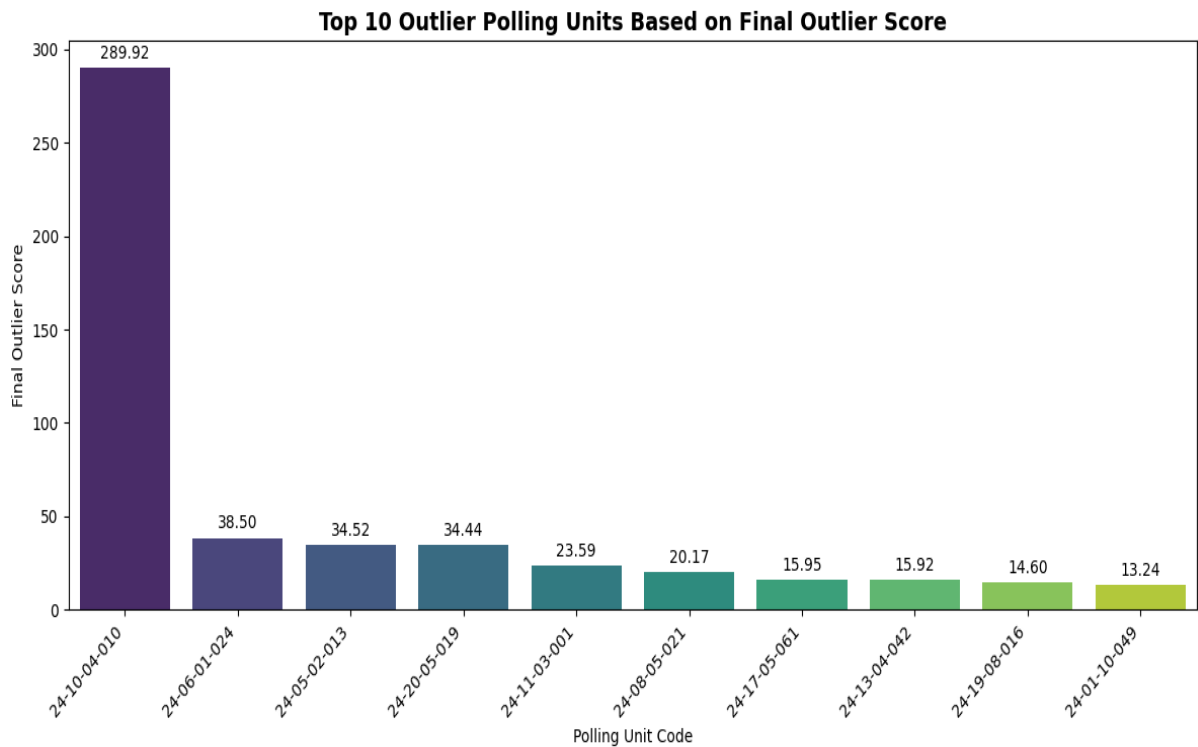
- **LP:** 523,882 votes (47.5% of total votes)
- **APC:** 502,947 votes (45.6% of total votes)
- **PDP:** 69,700 votes (6.3% of total votes)
- **NNPP:** 7,114 votes (0.6% of total votes)

Vote Share by Party in Lagos State Election



Top 10 Overall Outliers (Across All Parties)

Rank	LGA	PU-Code	LP	APC	PDP	NNPP	Outlier Score
1	IFAKO-IJAYE	24-10-04-010	380	240	312	255	289.92
2	BADAGRY	24-06-01-024	518	1	554	9	38.5
3	APAPA	24-05-02-013	36	1	44	50	34.52
4	SURULERE	24-20-05-019	4	67	1	49	34.44
5	IKEJA	24-11-03-001	1	68	5	27	23.59
6	ETI-OSA	24-08-05-021	539	0	124	0	20.17
7	OJO	24-17-05-061	0	30	1	43	15.95
8	KOSOFE	24-13-04-042	112	61	73	1	15.92
9	SOMOLU	24-19-08-016	1	64	4	17	14.6
10	AGEGE	24-01-10-049	4	61	0	23	13.24



Detailed Analysis of Top 3 Outliers

1. Top Outlier: IFAKO-IJAYE LGA (PU: 24-10-04-010)

Anomaly Score: 289.92

Vote Distribution:

LP: 380 votes

APC: 240 votes

PDP: 312 votes

NNPP: 255 votes

Analysis:

- This polling unit shows extremely high voter participation across all parties.
- Total votes (1,187) significantly exceed typical polling unit patterns.
- All parties received substantial votes, unlike neighbouring units where typically 1 or 2 parties dominate the voting system.
- **Possible explanations:** Data entry error, special voting centre, or genuine exceptionally high voting turnout.

Geospatial Context: Located in Ifako-Ijaye LGA, this polling unit's voting pattern deviates dramatically from surrounding polling units where vote totals are typically in the 100-300 range.

2. Second Outlier: BADAGRY LGA (PU: 24-06-01-024)

Anomaly Score: 38.50

Vote Distribution:

- LP: 518 votes
- APC: 1 vote
- PDP: 554 votes
- NNPP: 9 votes

Analysis:

- The data indicates extreme partisan polarization, with APC receiving near-zero votes notwithstanding its prominence as a major party.
- Very high votes for both LP and PDP, unusual competitive pattern.
- Total votes of (1,082), which is exceptionally high.
- Suggests either strong local political dynamics or potential data irregularity.

Geospatial Context: In Badagry LGA, this unit stands out for both the complete absence of APC support and the unusually competitive LP vs PDP race.

3. Third Outlier: APAPA LGA (PU: 24-05-02-013)

Anomaly Score: 34.52

Vote Distribution:

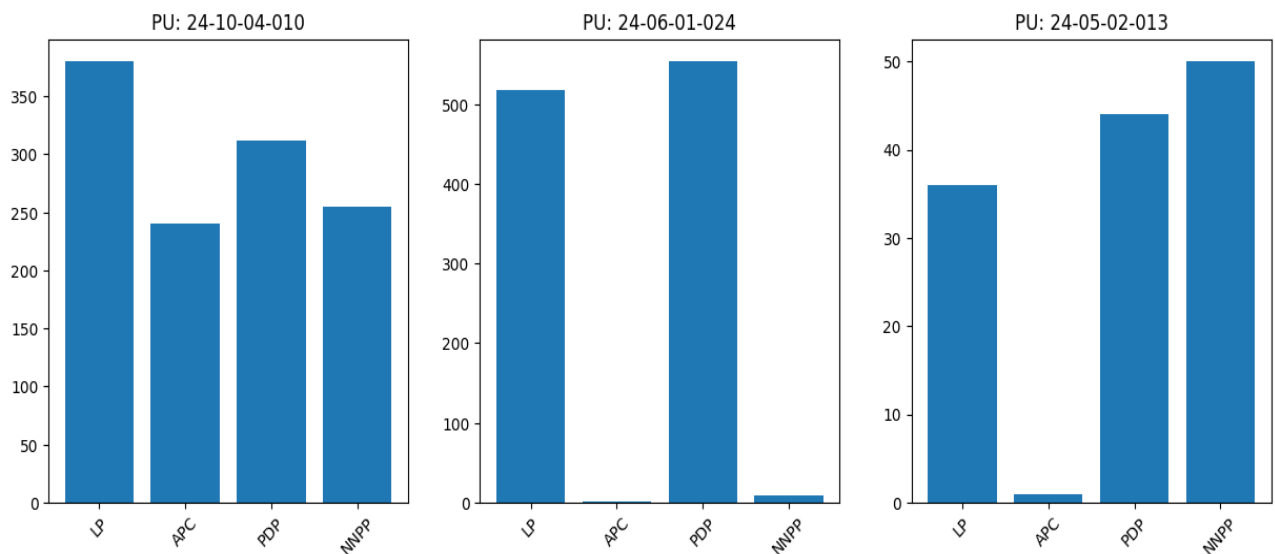
- LP: 36 votes
- APC: 1 vote
- PDP: 44 votes
- NNPP: 50 votes

Analysis:

- **NNPP** dominance is highly unusual as NNPP typically receives minimal votes statewide (0.6% average).

- Very low APC support similar to second outlier.
- Moderate total votes (131) but unusual party distribution.
- Could indicate local NNPP stronghold or data reporting issue.

Geospatial Context: In Apapa LGA, in this unit, NNPP suddenly does way better than in the nearby units, where they usually get almost nothing, like 0 to 5 votes. That just doesn't fit the usual pattern.



4. Party-Specific Outlier Patterns

LP Outliers

In the identified polling units, people either go all-in for the Labour Party or ignore it completely. There's rarely any middle ground. You get either huge number for LP or none at all, and not much in between. This kind of split points to certain areas where the party really takes over, or places where it just doesn't connect with voters. Most of the outliers are spots where LP support shoots way past what's normal, showing clear pockets where the party holds serious sway, maybe even strongholds across the electoral landscape.

APC Outliers

These polling units are marked by either a complete lack of APC votes or unusually strong APC performance. In many cases, APC received only 0–1 votes

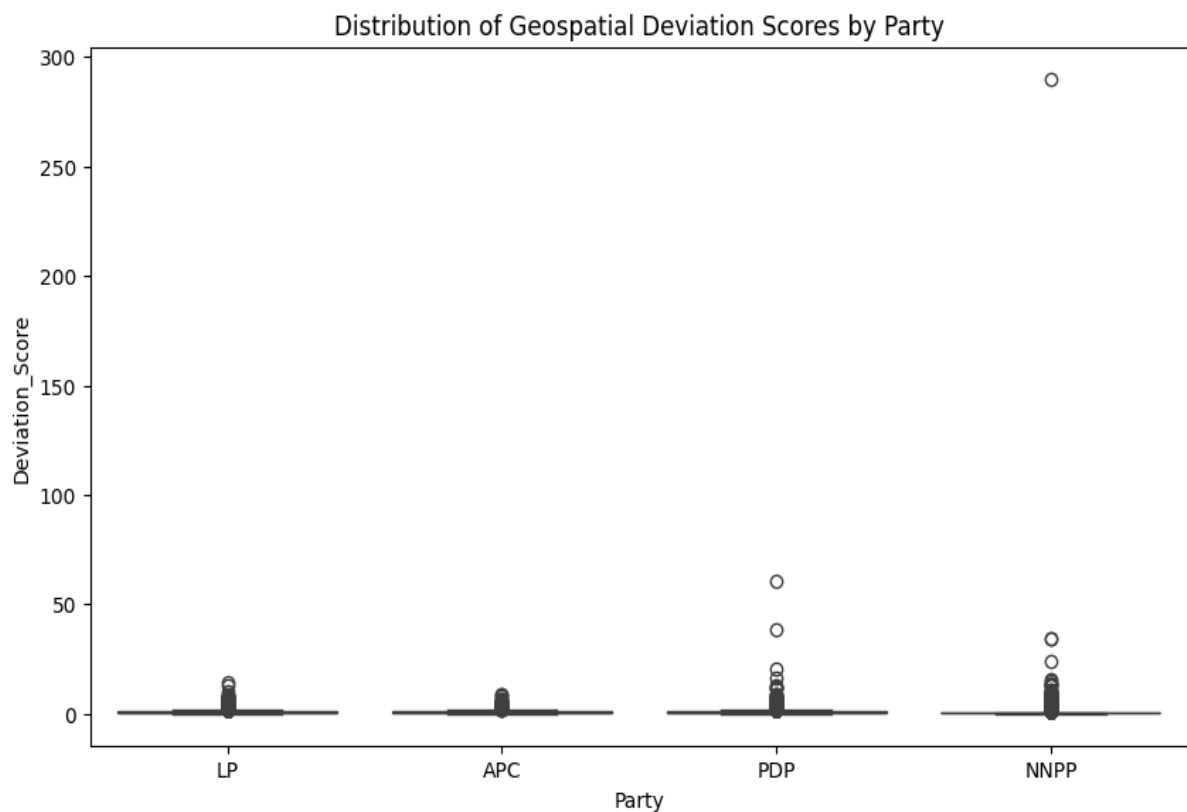
in areas that were otherwise competitive, suggesting localized irregularities or unique voting dynamics that deviate from the broader regional trend.

PDP Outliers

These polling units typically display PDP performance significantly above the state average. In several cases, PDP secured vote counts exceeding the combined totals of LP and APC, suggesting localized surges in support or potential reporting inconsistencies relative to regional voting trends.

NNPP Outliers

The most pronounced deviations are associated with NNPP, primarily due to its typically low baseline vote share across the state. Outlier polling units frequently record NNPP receiving 20 or more votes in areas where neighboring units report only 0–2 votes, indicating isolated pockets of unusually high support or potential data inconsistencies.



Conclusion and Key Insights

Main Findings

1. Spatial Clustering

Outliers are not distributed randomly across the state. They show clear geographical clustering, with certain LGAs and wards having a higher concentration of unusual polling units. This pattern suggests that irregularities may be influenced by regional factors rather than just random statistical noise.

2. Partisan Polarization

Several polling units exhibit extreme partisan polarization. One party receives overwhelming support while major competitors get almost no votes. These sharp imbalances differ significantly from expected local voting patterns and may indicate localized electoral distortions or unique political dynamics.

3. Turnout Anomalies

A subset of outliers shows total vote counts that seem mathematically inconsistent with the number of registered voters or with surrounding polling units. These anomalies raise concerns about possible over-reporting, aggregation errors, or issues with voter roll data.

4. Data Quality Concerns

The presence of extreme outlier scores across multiple LGA's points to potential data quality issues. These may come from transcription errors, data entry inconsistencies, or problems in the original result collation process. It is recommended to verify these findings against source documents, such as Form EC8A.

Recommendation for Further Investigations

- Physically verify results from the top 10 most anomalous polling units.
- Cross-check findings against election observer or incident reports.
- Audit for transcription or data entry errors in the original result sheets.
- Analyse spatial clustering of outliers across different LGAs to identify regional trends.

Conclusion

The geospatial outlier detection was able to detect Polling Units with statistically unusual voting pattern. The top outliers show combinations of extremely high turnout, complete party absences, and unusual party performance distributions that warrants further investigation for either understanding the dynamics of local politics or identifying data quality issues in the electoral results.

The method provides a large framework for election anomaly detection that could be used in future elections with proper parameter tuning and validation.